

Socioeconomic Impacts of Terrestrial Protected Areas: Evidence from Large-Scale National Surveys in Madagascar

Abstract Protected areas are the most widely used instrument for biodiversity conservation, yet their socioeconomic effects on nearby populations remain contested, particularly in low income contexts where livelihoods depend heavily on natural resources. This paper evaluates the impact of terrestrial protected areas on rural household well being in Madagascar, exploiting geolocated socio-demographic surveys covering the period of rapid protected area expansion between 2008 and 2021. To support identification, we draw on an earlier survey wave from 1997 to assess the plausibility of parallel trends prior to treatment. We combine spatially derived socio environmental indicators with genetic matching and difference in difference to construct a credible counterfactual. The study follows a pre analysis plan registered before any outcome analysis, including an ex ante power calculation indicating the ability to detect effects of approximately 7.5 wealth percentiles. We find no statistically significant average effect of protected area creation on the wealth index of households located within 10 km of protected areas established after 2008. A series of robustness checks using alternative data sources and specifications supports this result. The absence of an average effect may reflect offsetting positive and negative mechanisms, heterogeneous impacts across households or sites, or limited implementation capacity in a context of weak governance and chronic underfunding of protected areas.

Keywords: Biodiversity Conservation, Well-being, Demographic and Health surveys, Genetic matching, difference-in-difference, Madagascar

JEL classification: Q57, I31, C31, Q56, O55

1 Introduction

The reconciliation between conservation and economic development has long debated in the scientific literature (Adams et al. 2004), but the issue has gained renewed prominence over the past decade with the rapid global expansion of protected areas (PA). This debate is particularly salient in the context of the Kunming–Montreal Global Biodiversity Framework, under which 195 signatory states committed to protecting 30 percent of terrestrial land by 2030.

In theory, PAs can affect local livelihoods through multiple and potentially opposing channels. While they are a central instrument for biodiversity conservation (Maxwell et al. 2020), their es-

tablishment may restrict access to land and natural resources that support income generating activities such as agriculture, hunting, fishing, or forest product collection. At the same time, protected areas may generate benefits through compensation schemes, employment opportunities linked to conservation or tourism, and the provision of ecosystem services such as water regulation, erosion control, or fire prevention (Kandel et al. 2022).

Despite these ambivalent mechanisms, rigorous quantitative evidence on the socioeconomic impacts of PAs remains limited, in particular in poor and fragile governance contexts. Among the 1,043 studies reviewed across 104 countries by McKinnon et al. (2016), only 19 used quantitative methods to assess effects on material living conditions or economic well being. More recent syntheses confirm substantial heterogeneity in estimated impacts across contexts and methodologies. Focusing on 30 quantitative evaluations of household income, Kandel et al. (2022) find that protected areas are associated with modest average gains, with effects that depend strongly on local conditions. This heterogeneity underscores the need for context specific evaluations using transparent and robust empirical strategies.

Madagascar provides a particularly relevant setting for such an analysis. It ranks among the poorest countries globally under SDG 1.1, with the highest share of the population living below the international poverty line (Conceição 2024, pp. 298-299). During the study period, terrestrial PA coverage expanded from 3.6 percent of national territory in 2008 to 10.8 percent in 2021, while the share of the population living within 10 km of a PA rose from 9 to 28 percent. At the same time, Madagascar exhibits low state capacity (Hanson and Sigman 2021), which constrains the implementation of conservation policies and associated social measures. Combined with high rural dependence on natural resources, these features suggest that the socioeconomic impacts of protected areas may differ from those observed in less precarious institutional contexts.

Yet national scale impact evaluations remain scarce. None of the quantitative studies reviewed by McKinnon et al. (2016) focus on Madagascar. The only study cited by Kandel et al. (2022) that includes the country relies on cross sectional municipality level data from the 1993 census (Mammides 2019), predates the major expansion of protected areas, and does not allow for before and after comparisons.

This article contributes to the literature in two ways. Empirically, it provides the first national level impact evaluation of terrestrial protected areas in Madagascar, covering 71 protected areas created between 2008 and 2021. Methodologically, it combines recent advances in matching and difference in differences methods with geolocated household survey data to construct a credible counterfactual. To enhance transparency and limit researcher discretion, the empirical strategy was fully specified in a pre analysis plan registered on the Open Science Framework prior to conducting any outcome analysis. Our analysis plan was submitted with a dated and verifiable certification on the OSF portal in November 2024 and updated in March 2025.

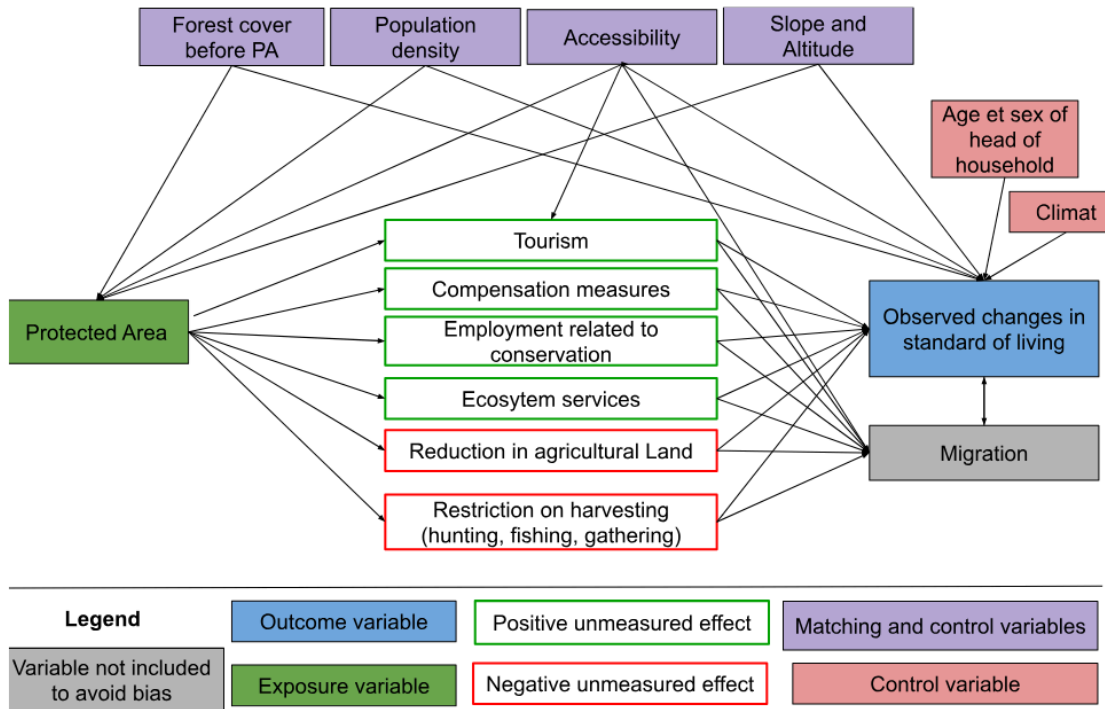
The remainder of the paper proceeds as follows. Section 2, presents the conceptual framework and hypotheses. Section 3 describes the data sources. Section 4 outlines the identification strategy and econometric methods. Section 5 presents the econometric approaches used to assess the effect of PA on household livelihoods, the effect of PA on inequalities between households, and the het-

erogeneity of effects according to the type of PA governance. Section 6 presents our main results and proposes a series of robustness tests. Section 7 concludes.

2 Theory of change

Our evaluation model is based on a theory of change that links the implementation of PA (the treatment) to local household well-being (outcome) through multiple possible channels, summarised in (Figure 1). The objective here is to determine the impact of PA implementation on observed changes in household living standards. Existing evidence suggests that average impacts are modest but highly context dependent. Kandel et al. (2022) report a slightly positive average impact, alongside substantial heterogeneity across settings. Building on this literature, we represent potential mechanisms in the form of directed acyclic graph (Hünermund and Bareinboim 2023). If the mechanisms represented affect all residents of a locality in a convergent manner, they should translate into a significant average impact (positive or negative) on the average well being (Hypothesis 1). If, on contrary, they affect households in very different ways, the average effects may be close to zero, but inequalities within local communities may increase (Hypothesis 2).

Figure 1: Logic diagram of the theory of change tested in the study



The factors likely to lead to a decline in well-being seem particularly significant in the Malagasy context, where the population is predominantly rural and living in extreme poverty (the last assessment was in 2012, with 80.7% of the population below the \$2.15 a day threshold at 2017 PPP). Six studies conducted in Madagascar between 1995 and 2006 estimated the opportunity cost of restricted access to natural resources following the creation of PAs (slash-and-burn agriculture, hunting, gathering, timber, etc.) at between USD 39 and 177 per household per year (Neudert,

Ganzhorn, and Waetzold 2017). Golden et al.(2014) estimated that income from hunting accounted for 57% of household's cash income in areas adjacent to the Makira and Masoala PA. Another survey of people living near Makira estimated the value of pharmaceutical use at USD 30-44 per year per household, based on the subsidized price of equivalent treatments in the Malagasy market (Golden et al. 2012).

Several factors that could help improve livelihoods through conservation appear to be fragile in Madagascar, starting with tourism. Naidoo et al. (2019) aggregate data from DHS surveys conducted between 2001 and 2011 in 34 developing countries. Their study is based on matching households near and far from PA, but with no pre-post conservation comparison. They highlight positive impacts, but only for a subset of PA 'with documented tourism'. According to their study, households living near the PA 'with tourism' are 17% wealthier and 16% less likely to be poor than similar households living far from these areas. However, tourism in Madagascar's PA remains low. According to data from Madagascar National Parks (MNP), only seven PA recorded more than 10,000 visitors in 2023 (with a maximum of 30,744 in Isalo), which is low compared to the average of 356,405 visitors per year and per PA recorded across 929 PAs worldwide in the global study by Chung et al. (2018).

When new PAs are created in Madagascar, compensation mechanisms for local populations remain rare, ineffective and insufficient (Rivière 2017; Bertrand et al. 2014). The most in-depth study on this subject, conducted by Poudyal et al. (2018) with support from the World Bank, focuses on the Ankeniheny Zahamena Corridor (CAZ), created in 2015 to connect several existing PAs. Five study sites were selected: two adjacent to the new CAZ PA (one eligible for compensation, the other not), two adjacent to long-established PA, and one far from the forest boundary. The median cost of the conservation restriction is estimated at USD 2,375 per household per year, representing between 27% and 84% of the average annual income. The amounts set aside to compensate beneficiary households were assessed to be insufficient relative to the losses incurred, and around 50% of households eligible for compensation received nothing (Mahesh Poudyal, Jones, et al. 2018; Poudyal et al. 2016).

Our first set of results therefore consists of determining whether PAs in Madagascar, by limiting access to natural resources, have negative impacts on the standard of living of households living nearby, potentially exceeding the benefits of compensation and ecosystem services, and whether these impacts are more adverse than those documented in other contexts (Hypothesis 1).

The impact mechanisms represented in Figure 1 are likely to affect households differently depending on their prior characteristics, which would increase inequality (hypothesis 2). Compensation measures are generally implemented in the form of projects to promote income-generating activities (agriculture, livestock, handicrafts) in surrounding communities (Mahesh Poudyal, Rakotonarivo, et al. 2018). In the context of such development projects, individuals known as "development brokers" frequently emerge as intermediaries between local communities and implementing organizations. By mobilizing their social networks and specific skills, these brokers manage to capture a disproportionate share of the benefits of interventions, whether in form of income or privileged access to opportunities. This dynamic can reinforce pre-existing inequalities within communities, limiting the access of the most vulnerable households to the expected benefits of

compensation programs. Although tourism development is often presented as an opportunity for economic growth, it also tends to exacerbate socioeconomic inequalities, particularly in developing countries. Adeniyi et al. (2024) show that in Southern Africa, tourism can initially exacerbate inequalities by concentrating benefits in the most attractive regions, while leaving marginalized communities out of the economic benefits. According to Ghosh and Mitra (2021), the relationship between tourism and inequality follows an inverted Kuznets curve in developing countries: when tourism remains moderate, its growth reduces inequalities, but when tourism becomes massive, further expansion worsens inequalities. Finally, Xuanming et al. (2024) point out that while tourism helps to improve certain socioeconomic indicators, it can also generate inflationary pressures and strain local resources, particularly affecting the most vulnerable households. Taken together, these mechanisms suggest that PAs may exacerbate economic inequalities within neighboring communities by creating opportunities that primarily benefit individuals with higher levels of education or dominant social positions, granting them access to rents and jobs related to tourism and associated activities (Hypothesis 2).

IUCN status of PAs¹ are frequently used to explain differences in effectiveness between them. For example, Naidoo et al. (2019) show that multiple-use PAs (statuses V and VI) tend to have more beneficial effects than strict areas (statuses I to IV), partly due to greater flexibility in integrating local needs. Beyond status alone, governance plays a central role. Eklund et al. (2017) highlight the importance of transparent and inclusive structures to maximize the positive effects of PAs on conservation and social justice. They call for management approaches to be adapted to local contexts, with greater involvement of communities in decision-making processes, to better reconcile conservation and development objectives.

This diversity is particularly evident in Madagascar. Although governed by similar formal statuses, PAs follow different paths depending on the local context and the way in which they are implemented. Froger and Méral (2009) show that the early initiatives of shared governance, gradually introduced with in-depth mediation efforts, achieved encouraging results by strengthening local community support. However, from the 2000s onward, the accelerated deployment of management transfers, driven by quantitative targets, often led to hasty and less contextually adapted implementations, undermining the effectiveness of these mechanisms. These experiences demonstrate that, beyond the PA status, their establishment period, management approach, and level of community participation significantly influence their socioeconomic impacts. We therefore anticipate that the impacts of PAs on well-being and inequalities are heterogeneous, and that PAs with higher levels of local community participation are more likely to generate greater benefits and distribute them more equitably (Hypothesis 3).

Based on this theory of change, we define two main outcome variables to capture changes in living standards: household wealth index (the main outcome) and the standardized Z-score of the wealth index (the secondary outcome). Household wealth will be the outcome variable used to determine the overall impact of PAs, and the standardized Z-score of the wealth index will capture inequalities in living standards across localities. We also use variables that may be predictive of

¹<https://www.google.com/url?q=https://portals.iucn.org/library/efiles/documents/PAPS-016-Fr.pdf&sa=D&source=docs&ust=1768418520379026&usg=AOvVaw26wfaFnma2KaAr7cxRAz-5>

the outcome under study. The appropriate covariates for our model are variables that are likely to influence both the probability of treatment (whether a PA has been created near the household) and the outcome (household living standard and inequalities between households). The literature shows that PAs tend to be created in less dense, less accessible, higher and steeper regions (Joppa and Pfaff 2010). These variables may also affect living standards: areas that are more dense, flat, low-lying and accessible (in terms of travel time and geography) tend to be wealthier (Gallup, Sachs, and Mellinger 1999). We propose five covariates (forest cover in 2000, slope, elevation, population density in 2000, and accessibility in 2000).

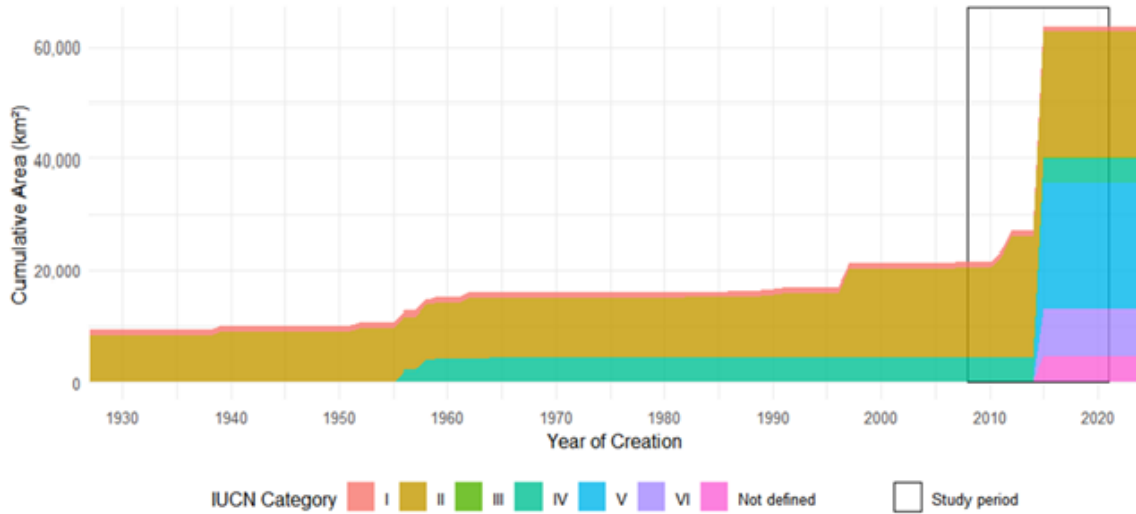
3 Data

Considering the long-term, large-scale, complex, and politically sensitive nature of the intervention to be evaluated, we use secondary data on the socioeconomic conditions of households, their geographical environment and their location in relation to PAs.

3.1 Protected areas

This study evaluates the impact of terrestrial PA creation on rural household well-being between 2008 and 2021. These time frame was chosen on the basis of the availability of geolocalised data on household living conditions and coincides with a period of strong expansion of PAs in the country, as shown in Figure 2

Figure 2: Evolution of protected areas in Madagascar and study period

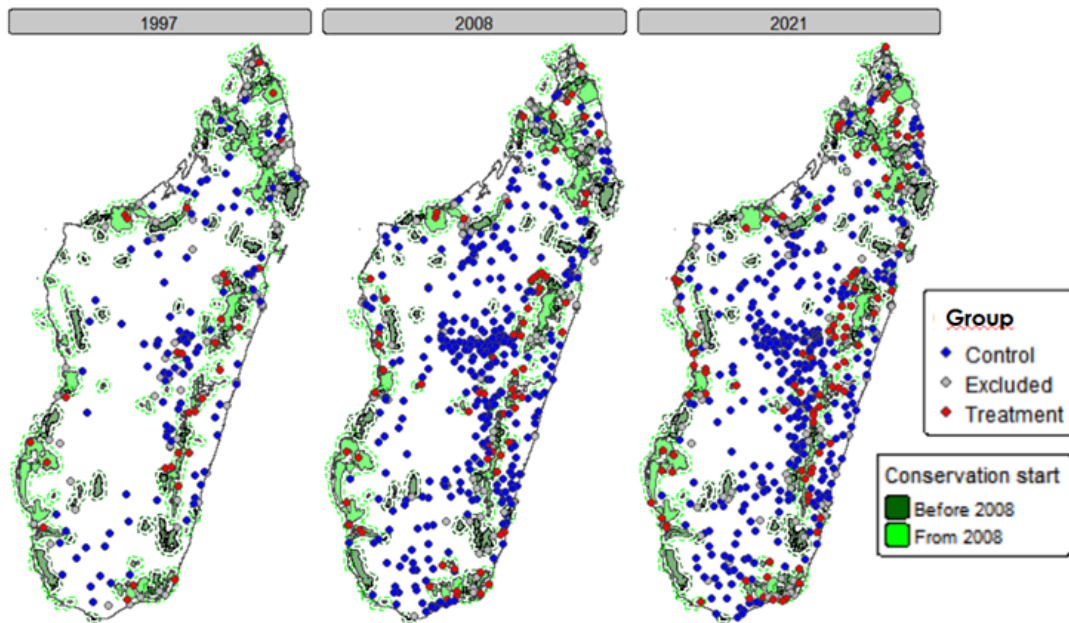


Notes: This graph shows the evolution of PA creation in Madagascar since their creation in 1927 (under the colonial administration) until 2024. From 1927 until the early 2000s, PAs were characterized by strict conservation (IUCN categories I, II, and IV). At the IUCN Parks Summit in Durban in 2003, the Malagasy government committed to tripling coverage, which led to the creation of new PAs with 28 provisional creation decrees published between April 2006 and December 2007 and a global decree bringing the number of new PA to 97 in 2008. Final ratification

decrees were not issued until 2015, which formalized these creations of new PAs. The data used to produce the graph come from the Service de la Gouvernance des Aires Protégées (SGAP), of the Ministère de l'Environnement et du Développement Durable (MEDD)

Our study analyzes the impact of PA on household well being over 13 years (2008-2021). We use 2008 as the reference year. Accordingly, the population considered as treated encompasses households living in rural areas within 10 km of a PA created between 2008 and 2021, according to the GPS coordinates provided in the Demographic Health Surveys (DHS) data². Households in the control group are those living in a rural area more than 10 km away from a PA created between 2008 and 2021, and they exhibit very similar characteristics or share significant traits with households in the treatment group. We decided to exclude rural populations living within 10 km of PA created before 2008, as they are considered treated before the study period; as well as households located in urban areas.

Figure 3: Classification of household clusters according their group affiliation



Notes: Households in the treatment group are those located in clusters less than 10 km from PAs created since 2008 in rural areas (shown in red) and households are those located in clusters more than 10 km from PA created since 2008 in

²These GPS coordinates correspond to the centroids of the enumeration areas surveyed. To protect respondent confidentiality, these coordinates are first randomly shifted using the following procedure: An offset angle between 0 and 360 is randomly drawn, and then an offset distance is randomly drawn, between 0 and 2 km in urban areas and between 0 and 5 km in rural areas. For 1% of rural clusters, the distance drawn is between 0 and 10 km (Skiles et al. 2013)

rural areas (shown in blue). Rural populations living within 10 km of PA created before 2008 or located in urban areas are excluded from the study (shown in black). The GPS data used comes from DHS clusters.

Table 1 shows the distribution of terrestrial PAs (by number and area) according to the period in which they were created by final decree, taking 2008 as the reference year. In the treatment period (2008 to 2021), 103 PAs were created, covering 75,252 km².

Table 1: Classification of household clusters according their group affiliation

Protected Areas by period (Reference year = 2008)		
terrestrial areas only		
Creation Period	PA number	Area (km ²)
Before 2008	60	47,160
Starting in 2008	84	38,983

Notes: PA number” refers to the number of protected areas created during each period. Area is expressed in square kilometers (km²). Calculations are based on data from the World Database on Protected Areas (WDPA).

3.2 Household socioeconomic conditions

The data on household living conditions used for this study comes from surveys conducted by the “Institut National de la Statistique de Madagascar” (INSTAT) as part of the Demographic and Health Surveys (DHS)³. These data covers a wide range of topics, including demographic characteristics, living conditions, health, education, sanitation, and hygiene. These surveys were conducted 1997, 2008, and 2021, containing 650 clusters in 2021⁴, 585 in 2008⁵, and 268 in 1997⁶.

³The DHS surveys are based on a two-stage stratified sampling method. The population of interest is divided into 23 study areas corresponding to Madagascar’s 22 regions, the capital Antananarivo(considered separately), an the Analamanga region without the capital (to isolate the impact of the capital on regional results). With the exception of the capital two strata were created in each study area. At the first level, enumeration areas (also called ‘clusters’) are randomly selected within each domain, with a probability proportional to the population of the cluster according to the latest census. At the second level, a sample of households is randomly selected within these clusters to participate in the survey programs

⁴657 clusters were drawn with probability proportional to size. After implementation in the field, 650 of the 657 clusters initially selected were actually visited

⁵600 clusters were drawn with probability proportional to size. Of the 600 clusters selected, 596 could be surveyed. However, nine other clusters had invalid GPS coordinates, resulting in a total of 585 clusters for 2008

⁶270 clusters were drawn with probability proportional to size. Of the 270 clusters selected, 269 could be surveyed. However, one cluster had invalid GPS coordinates, resulting in a total of 268 clusters for 2008.

These data are used to construct the variables for the impact assessment model. In this analysis, two outcome variables are considered: household living standards (primary outcome) and inequalities in living standards at the level of the surveyed localities (secondary outcome).

- Main outcome variable: Household living standards

The first outcome variable, household living standards, is estimated from the wealth index, calculated specifically for rural areas (variable coded hv270a in the DHS data). The wealth index is defined in the DHS data catalogue as: ‘A composite measure of a household’s cumulative living standard. The wealth index is calculated using easy-to-collect data on a household’s ownership of selected assets, such as televisions and bicycles; materials used for housing construction; and types of water access and sanitation facilities. Generated with a statistical procedure known as principal components analysis, the wealth index places individual households on a continuous scale of relative wealth. DHS separates all interviewed households into five wealth quintiles to compare the influence of wealth on various population, health and nutrition indicators. As a response to criticism that a single wealth index is too urban in its construction and not able to distinguish the poorest of the poor from other poor households, this variable provides an urban- and rural-specific wealth index’ (The DHS Program/ICF 2018). As described above, we will translate this wealth index into an integer between 1 and 100, corresponding to the household’s wealth percentile relative to the distribution of the whole sample.

- Secondary outcome variable: inequality of household living standards In addition to the impact of PAs on household living standards, we seek to understand their influence on socioeconomic inequalities within the affected populations. To do this, we propose to use a standardized Z-Score of the wealth index, allowing for the comparison of the relative distribution of wealth around the mean within the study population, at the level of each survey cluster.

The Z-Score Z_i for each household i is calculated from the wealth index using the following formula:

$$Z_i = \frac{W_i - \mu_W}{\sigma_W}$$

(1)

where W_i is the wealth index for household i , μ_W represents the average wealth index for all households surveyed in each rural cluster, and σ_W is the standard deviation of households in the cluster.

- Control variable These data also provide us with control variables: age and sex of the head of the household. These characteristics determine the economic opportunities and resource management capacity of households (Lo Bue et al. 2022). These variables reduce the unexplained variability in the model and thus enable a more accurate estimate of the treatment effect.

3.3 Household geophysical environment

Household geophysical environment data will be obtained using the R package `mapme.biodiversity` (Görge and Bhandari 2022). This package automates the retrieval and processing of large

raster-format data to produce a series of indicators applied to user-defined polygons for specified periods, which are subsequently and defined.

“Forest cover rates in 2000 correspond to the percentage coverage by vegetation with a height of 5 meters or more (Hansen et al. 2013). This variable is provided by the Global Forest Change dataset, which indicates for each pixel of one arc-second (approximately 30 meters at the equator) an estimate of forest cover in 2000 (Hansen, Potapov, Moore, Hancher, Turubanova, Tyukavina, Thau, Stehman, Goetz, Loveland, and others 2013). In Madagascar, as in other countries, PA have been preferentially created by targeting forest zones (Wilson et al. 2006; Carvalho et al. 2020). Globally, Naidoo (2019) indicates that forest cover in an area is negatively correlated with the living standards of its inhabitants.”

“Slope and elevation are calculated using a digital terrain model from NASA-SRTM (Shuttle Radar Topography data), which provides an elevation estimate for arc-second pixels (NASA JPL 2020). Slope is measured as a percentage for each plot, while elevation is measured in meters. These topological variables influence the location of PA (Joppa and Pfaff 2010), as well as the agricultural potential of an area and therefore the living standards of rural populations (Canavire-Bacarreza and Hanauer 2013).”

“Population density in 2000 corresponds to the estimated number of inhabitants per km² based on Worldpop data. Worldpop data provide estimates of population density for the year 2000 at a spatial resolution of about 1 km. They use modeling techniques and combines census data with various geospatial datasets(WorldPop and CIESIN 2018).”

“Accessibility in 2000 corresponds to the estimated travel time for households to the nearest cities, measured in minutes. These accessibility data to cities are compiled by the Joint Research Center (JRC), with 2000 as the reference year (Uchida and Nelson 2011). Accessibility to cities determines the ability to benefit from the services, products and opportunities they offer and is therefore a key factor in determining living standards in rural areas (Weiss et al. 2018; INSTAT 2020).”

Each of these variables will be calculated within a circle of 10 km radius around the GPS coordinates of the survey cluster. All households in the same cluster are assigned the same values for forest cover rate in 2000, slope and elevation, population density in 2000 and accessibility in 2000.

We also integrate the Standardized Precipitation Evapotranspiration Index (SPEI) (2010) for the year preceding the survey. This index is calculated based on a long-term reference (1981-2010) to quantify excess or deficit of rainfall. We use the SPEI value for the two years prior to each survey, as well as the year of the survey. The SPEI is calculated from monthly precipitation and minimum and maximum temperature data from worldclim, using an improved version of the Hargreaves method defined by (Droogers and Allen 2002), as implemented in the SPEI R package (2010). This variable will be calculated for a circle with a radius of 10 km around the GPS coordinates of the cluster.

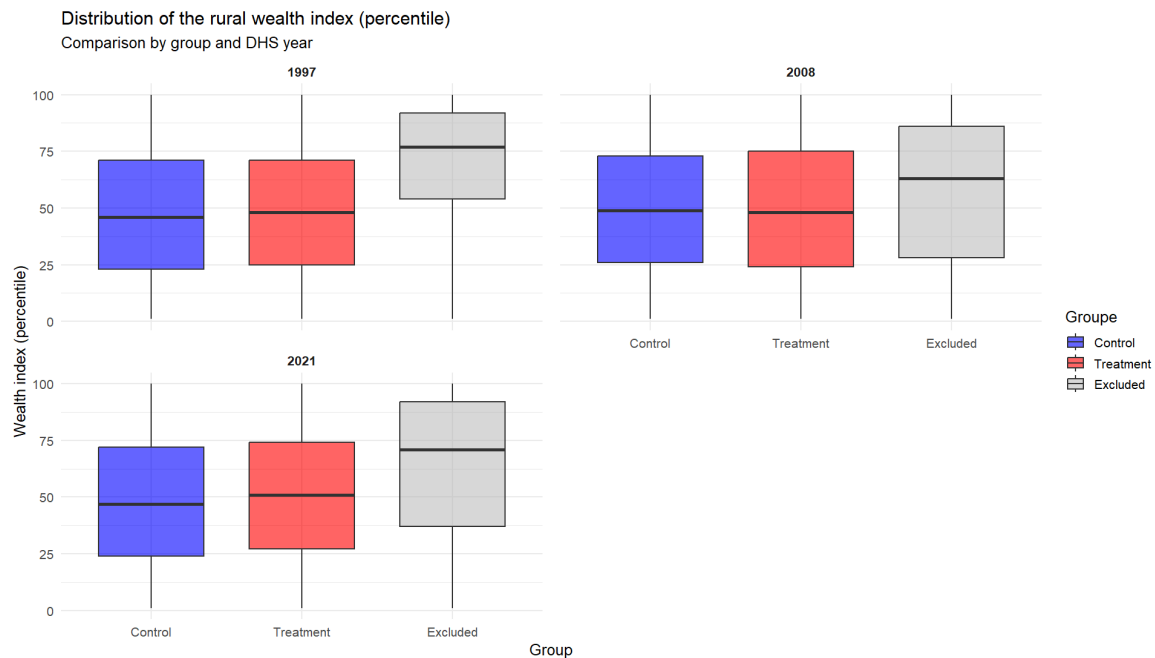
The data from this source will constitute part of our matching variables. Matching covariates are used to select units not exposed to conservation (control groups) that are comparable to the exposed units (treatment group). A appropriate for the matching process are variables likely to

influence both the probability of treatment (whether a PA has been created near the household) and the outcome (household living standard and inequalities among households).

4 Descriptive statistics

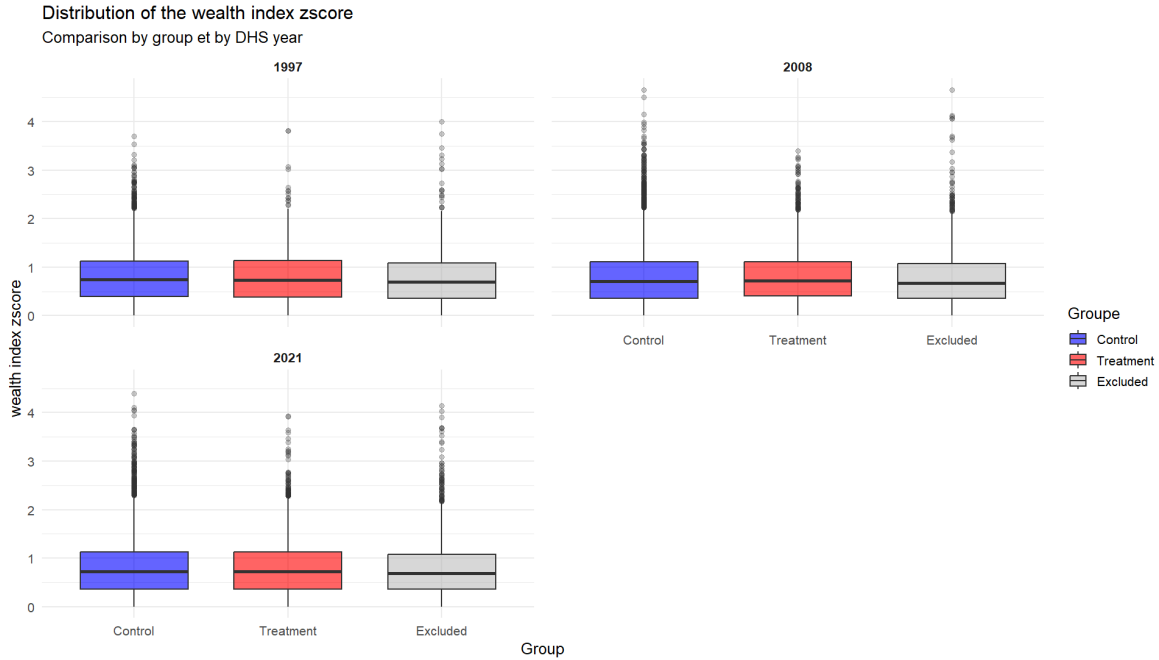
Figure 4 shows that the control and treatment groups have very similar wealth profiles, with averages between 47 and 50, and medians ranging from 46 to 51. In contrast, the excluded group is at the top of the national distribution of wealth, with an average above 50 and a median between 63 and 77. Its distribution is generally shifted upwards with higher values across the distribution. This suggests that there is no difference between the control and treatment groups; however, the standard deviations of the two groups indicate a high degree of heterogeneity in household living standards.

Figure 4: Distribution of the rural wealth index (percentile)



Notes: The graph shows boxplots of the wealth index distribution in percentiles (blue for the control group, red for the treatment group, and grey for the excluded groups). Across all surveys, for all years of the study, the wealth index distributions of the treatment and control groups are roughly similar.

Figure 5: Distribution of the Zscore of the rural wealth index (percentile)



Notes: The graph shows boxplots of the zscore wealth index distribution in percentiles (blue for the control group, red for the treatment group, and grey for the excluded groups). Across all survey years, the wealth index Z scores of the treatment and control group is roughly similar.

5 Empirical strategy

5.1 Matching methods

Our evaluative approach is based on a counterfactual measure that estimates the causal effect of the treatment, namely the implementation of PAs. This approach is based on a comparison between a treatment group (households living in rural areas within or less than 10 km of PA) and a control group (households living in rural areas more than 10 km from PA created after 2008) (Schleicher et al. 2020; Desbureaux 2021). The study thus fits within the framework of Rubin's causal model (Rubin 1974), according to which potential outcomes depend on exposure to the treatment. To ensure comparability between groups, we apply genetic matching (function GenMatch from the R MatchIt package) to pair each unit in the treatment group with a unit in the control group that has the same observable characteristics (Ho et al. 2007). This approach uses a single distance measure 'Mahalanobis distance', to quantify the similarity between the two groups of observations, while taking into account the correlations between covariates and their covariances (Diamond and Sekhon 2013). This procedure increases the credibility of the results and reduces endogeneity issues (Ma et al. 2020). We also apply a caliper of 0.25 standard deviation of the Mahalanobis distance to avoid creating pairs with overly large differences (Rosenbaum 1983). The validity of this estimation depends on the balance between the treatment and control groups

before and after matching. Before matching, we check the balance using the standardized mean difference (SMD), which measures the difference between the means of the covariates in the two groups. After matching, matching is further assessed visually using a quantile-quantile plot.

5.2 Difference-in-difference

We assess the impact of PAs on household living standards and inequalities using the difference-in-difference (DID) method. The DID principle consists in comparing the wealth index of control and treatment households before and after the establishment of PAs(our treatment begins in 2008).

This method relies on the parallel trends assumption, according to which treated and untreated areas evolved similarly prior to the intervention. To validate this hypothesis, we will use as a reference, among the rural households surveyed in 1997, those living within 10 km of a PA created between 2008 and 2021 (placebo treatment group) and those matched to them using the method described above (placebo control group). We will conduct a placebo test by performing a DID estimation between these groups for the period 1997-2008. If the result of this placebo test close to zero or statistically insignificant, the hypothesis of parallel trends is supported.

In our DID estimates, we use cluster robust standard errors to account for intra-cluster correlations.

The simplified equation for the difference-in-difference (Daw and Hatfield 2018) is as follows:

$$Y_{it} = \alpha_0 + \beta_1 \text{Treatment}_i + \beta_2 \text{Post}_t + \delta D_{it} + \varepsilon_{it}$$

(2)

With the parameters defined as:

Y_{it} : Wealth index for household i in year t (1997, 2008 or 2021) α_0 : Intercept, representing the baseline wealth index β_1 : Coefficient associated with the variable Treatment_i

Treatment_i : Binary variable that takes the value of 1 for households i affected by a PA (treatment households) and 0 for households i not affected by PA (control households)

β_2 : : Coefficient associated with the variable Post_t

Post_t : Binary variable that takes the value of 1 for the post-treatment year 2021 and 0 for the pre-treatment year 2008

δ : Coefficient of interest representing the effect of PA creation on household living standards i in year t

D_{it} : Interaction term between Treatment_i and Post_t ; It is equal to 1 for treatment households i after the treatment year t and 0 for the other households.

ε_{it} : Error term for household i in year t

5.3 Robustness and sensitivity tests

We implement a series of robustness tests predefined in the pre-analysis plan published prior to the results(published on OSF in March 2025:<https://osf.io/bgu5n/>).We re-estimate all effects using

distances thresholds 5 km and 15 km to assess the robustness of our results to the choice of the treatment radius. While confidence intervals may widen when restricting the study area to 5 km, comparing coefficients across specifications allows us to assess the consistency of the estimated effects.

We apply Benjamini-Hochberg's (Benjamini and Hochberg 1995) False Discovery Rate method to test Hypothesis 2 (PA impact on inequalities between households) and Hypothesis 3 (the role of PA governance model). These tests are performed to mitigate the risk of incorrectly inferring significant effects by controlling the average proportion of false positives among the results reported as significant. Hypothesis 2 is evaluated using the Z-score outcome of the wealth index, and Hypothesis 3 is evaluated using PA governance categories based on IUCN status of PAs.

In the analysis, the outcome variable may be correlated with unobserved factors or shocks at the household level. Fixed effects methods can correct for many of these factors, but only repeated cross-sectional data are available.

Genetic matching and Doubly Robust DID estimation on a cross-section do not fully control for unobserved characteristics that may simultaneously affect PA and the outcome variable (wealth index). To assess the robustness of the results in the face of these potential biases resulting from unobserved confounding variables, we perform a sensitivity analysis using Rosenbaum's method (2002).

6 Results

6.1 Matching between treatment and control

The distribution analysis of covariates prior to matching shows a substantial imbalance between the treatment and control groups (see Table). After matching, the SMDs improve for all variables in each year. However, the population density in 2000 remains the most difficult variable to balance. This reflects the fact PAs are not randomly distributed.

6.2 Overall impact on livelihoods

In our DID estimation, we used robust standard errors. The results indicate no statistically significant difference at the 5% level between treated and control households. Figure 6 presents the results of the 2x2 DID estimation comparing the treatment groups (households living within 10 km of a PA) and the control groups (households more than 10 km away) prior to (1997-2008) and after (2008-2021) the PA establishment.

Table 2

	m_pre		m_main	
	Placebo 97-08		Traitement 08-21	
Dependent Var.: wealth_centile_rural_weighted wealth_centile_rural_weighted				
Constant	48.63***	(4.599)	43.49***	(2.815)
treat	-2.140	(4.433)	1.227	(3.590)
post	-4.717	(6.369)	19.09***	(4.760)
treat_post	3.904	(5.699)	-2.570	(4.492)
spei_wc_n_2	3.138	(4.177)	-5.552	(3.496)
spei_wc_n_1	-1.337	(2.762)	-8.845***	(2.314)
spei_wc_n	2.997	(4.556)	18.88***	(2.656)
hv219Femme	-1.825.	(0.9782)	-3.460***	(0.7927)
hv220	0.1015***	(0.0272)	0.0957***	(0.0194)
S.E.: Clustered		by: hv001	by: hv001	
Observations	6,831		11,686	
R2	0.00523		0.07427	
Adj. R2	0.00406		0.07364	

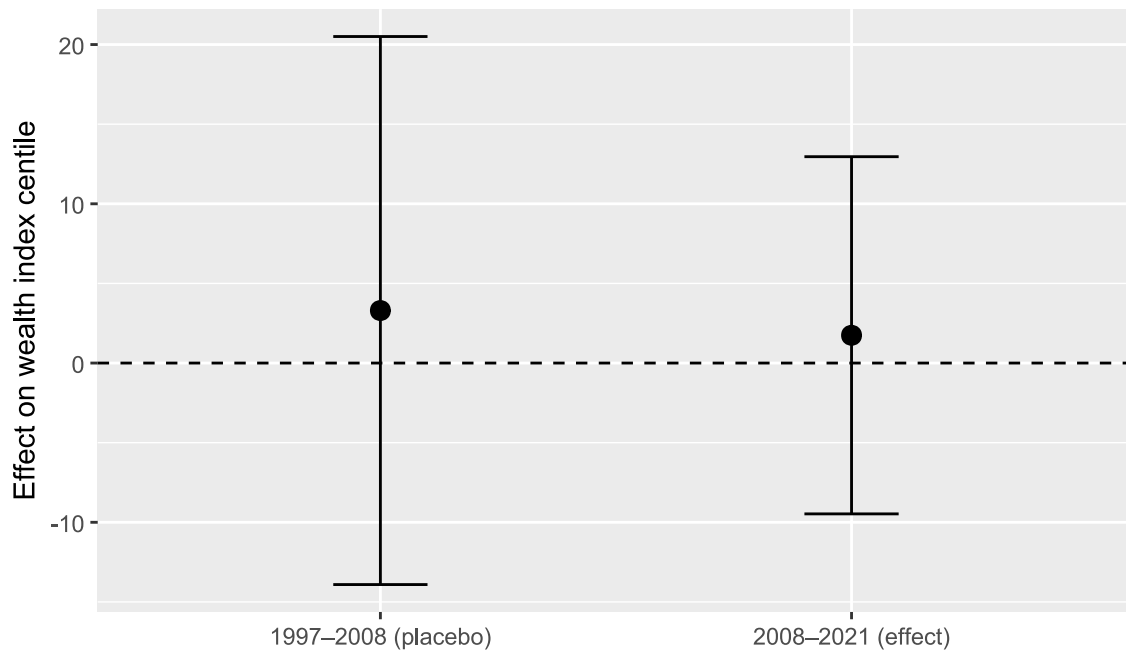
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1				

The placebo test for the treatment period (1997-2008 period) yields small and insignificant coefficient (+3.9 percentile) indicating that no differential trend is detected prior to intervention (Table 3). The supports the parallel trends assumption as the wealth indices of the two groups vary similarly before PA creation.

Climatic conditions, as measured by the SPEI have no significant effect on rural wealth, and the effect of household characteristics appears to be minimal. On average, households headed by women are slightly poorer, while those with older members are tend to be wealthier.

For the treatment period (2008-2021), the estimated effect is negative but statistically insignificant (-2.5 wealth percentiles). Households living near PAs appear to have slightly lower living standards, but the confidence interval clearly includes zero. Given the study's ex ante power calculation, which indicates that effects smaller than approximately 7.8 wealth percentiles cannot be reliably detected. This result that any true effect is either close to zero or below our detection threshold.

Figure 6: Impact on well-being: 2x2 DiD impact on local population (< 10km)



Notes: The vertical axis reports the estimated DID effect on the household wealth percentile (in points). Points indicate point estimates, and error bars represent cluster robust confidence intervals. The placebo estimate (1997-2008) and treatment period estimate (2008-2021) are both statistically insignificant, with the confidence interval crossing zero.

6.2.a Event study estimation

The 2×2 DID estimates indicate no statistically significant average effect of PA creation on household living standards. Consistent with the pre analysis plan, our empirical design is powered to detect effects of approximately 7.5 wealth percentiles, and no such effect is detected. To further explore the possibility of heterogeneous effects over time, we therefore apply an estimator adapted to staggered treatment adoption and multiple periods, which allows treatment effects to vary dynamically (Borusyak, Jaravel, and Spiess 2024).

To increase temporal coverage before and after treatment, we integrate additional survey waves from the Malaria Indicator Surveys (MIS) (1997, 2011, and 2013) and MICS 2018, which follows a survey design comparable to DHS and MIS and includes 17,870 households. These additional surveys improve coverage of both pre treatment and post treatment periods.

- Staggered DiD: static

Table 3: Static treatment effect estimates on well-being

	did2s (static)
treat_on = 1	4.519
	(4.234)
Num.Obs.	4246
R2	0.006
R2 Adj.	0.006
AIC	41025.8
BIC	41032.2
RMSE	30.32
Std.Errors	Corrected Clustered (cluster_uid)

The static staggered DID estimate suggests a positive but statistically insignificant association between treatment and household wealth (+4.5 wealth percentiles, standard error 4.23). Given the magnitude of uncertainty, this estimate does not allow us to reject the null hypothesis of no effect, motivating a dynamic event study analysis.

- Staggered DiD: dynamic event study

For the event study, each observation is indexed by its relative year to treatment, defined as the difference between the survey year and the year of PA establishment for treated locations. This transformation allows outcomes to be compared along a common event time dimension across cohorts. Because surveys are conducted at irregular intervals and some relative years are represented by very few observations, we group relative years into two-year bins. Observations occurring more than ten years before or after treatment are further aggregated into open-ended bins, as shown in Table 4. This binning strategy improves the stability and precision of the estimated dynamic effects.

Table 4: Binning of relative years-to-treatment for the event study

Rel. year	Group	Clusters	Obs.	Bin	Clusters	Obs.
-18	Treat.	7	218	min to -9	18	552
-16	Treat.	2	55			
-13	Treat.	1	27			
-12	Treat.	1	30			
-11	Treat.	4	129			
-10	Treat.	1	31			
-9	Treat.	2	62			
-7	Treat.	23	696	-8 to -7	23	696
-6	Treat.	1	32	-6 to -5	3	93
-5	Treat.	2	61			
-4	Treat.	12	349	-4 to -3	14	413
-3	Treat.	2	64			
-2	Treat.	26	817	-2 to -1	28	878
-1	Treat.	2	61			
0	Treat.	12	368	0 to 1	27	846
1	Treat.	15	478			
2	Treat.	1	32	2 to 3	21	652
3	Treat.	20	620			
4	Treat.	3	89	4 to 5	13	411
5	Treat.	10	322			
6	Treat.	37	1,141	6 to 7	38	1,172
7	Treat.	1	31			
8	Treat.	14	449	8 to 9	16	500
9	Treat.	2	51			
10	Treat.	7	222	10 to max	27	851
11	Treat.	8	260			
13	Treat.	12	369			
NA	Ctrl.	282	7,064	NA	282	7,064

Figure 7: Impact on well-being: Event-study with 2-year bins

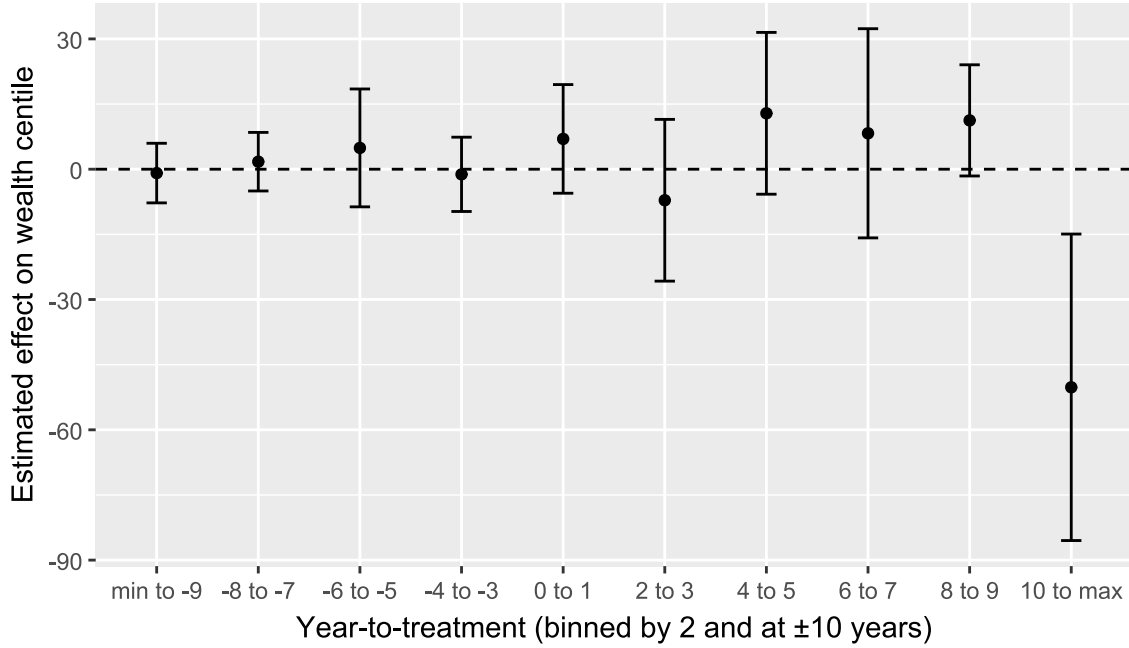


Figure 7 shows that the estimated event time coefficients are close to zero in the pre treatment period, and none of the pre treatment coefficients are statistically distinguishable from zero, supporting the plausibility of the parallel trends assumption. Following PA establishment, point estimates fluctuate across event time bins and are generally imprecise. However, the estimate for the longest horizon (more than 10 years after treatment) is large and negative, and its confidence interval lies entirely below zero. This pattern suggests a potential adverse effect emerging only at long horizons, while effects in the short and medium run remain statistically indistinguishable from zero. This long horizon estimate should nonetheless be interpreted cautiously, as it may reflect a specific subset of PAs. Although it includes 851 treated observations from 27 clusters, it primarily concerns PAs created at the beginning of the study period, which may differ systematically from later ones (for instance, lower tourism potential or other unobserved characteristics). Subject to further investigation, this estimate could reflect a genuine mechanism, such as a gradual decline in compensatory measures by PA managers or the cumulative effects of land use restrictions on younger farming generations.

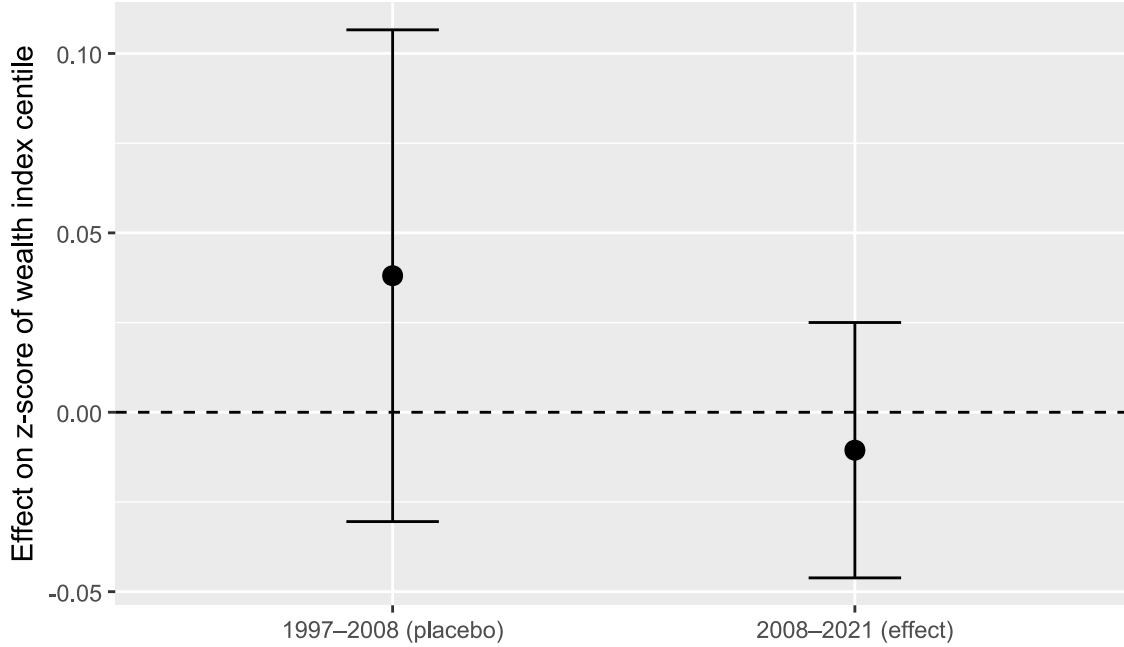
6.3 Effect on inequalities

6.3.a Estimation DID 2X2

The DID estimates indicate that PA creation does not have a statistically significant average effect on wealth inequalities, as measured by the Z score of the wealth index. The placebo estimation for the pre treatment period (1997–2008) yields a small and statistically insignificant treatment effect ($\text{treat} \times \text{post} = 0.038$, s.e. 0.035), supporting the plausibility of the parallel trends assumption for inequality outcomes. For the treatment period (2008–2021), the estimated treatment effect is negative but small and statistically insignificant ($\text{treat} \times \text{post} = -0.011$, s.e. 0.018). As shown in

Figure 7, the clustered confidence interval clearly overlaps zero, indicating no detectable causal effect of PA creation on within-cluster wealth inequality. Hypothesis 2 is therefore not supported in terms of an average effect.

Figure 8: Impact on inequalities: 2x2 DiD impact on local population (< 10km)



Notes: The estimated average effect (black dot) is slightly positive for the placebo period, but still not significant (the error bar crosses the horizontal line). During treatment, the estimated average effect is slightly below 0, pointing to a negative effect, but as the clustered confidence interval overlaps 0, the effect is not significant

The coefficient on the post indicator is positive and statistically significant in the main specification (post = 0.055, $p < 0.01$), indicating a general increase in inequality over time between 2008 and 2021 that affects both treated and control clusters alike. This pattern reflects a time trend rather than a treatment effect, and should not be attributed to PA creation. Climatic conditions, as measured by the SPEI, exhibit statistically significant associations with inequality in the treatment period, suggesting that climatic shocks disproportionately affect poorer households and contribute to widening disparities. In contrast, household demographic characteristics play a limited role: households headed by women exhibit slightly lower relative wealth positions in the inequality distribution, while the age of the household head is positively associated with relative wealth. Finally, the very low values of R^2 and adjusted R^2 indicate that the model explains only a small share of the variation in inequality outcomes. This is expected in short panel DID specifications using repeated cross sections and does not undermine the identification of the treatment effect, but it cautions against interpreting the model as predictive

- Staggered DiD: static specification

Table 5 reports the static staggered DID estimates for inequality outcomes. The estimated treatment effect is negative but very small (-0.007), and not statistically significant. This indicates that, on average, exposure to PA creation is not associated with a detectable change in within cluster wealth inequality. The estimate is imprecise, and the null hypothesis of no effect should not be rejected

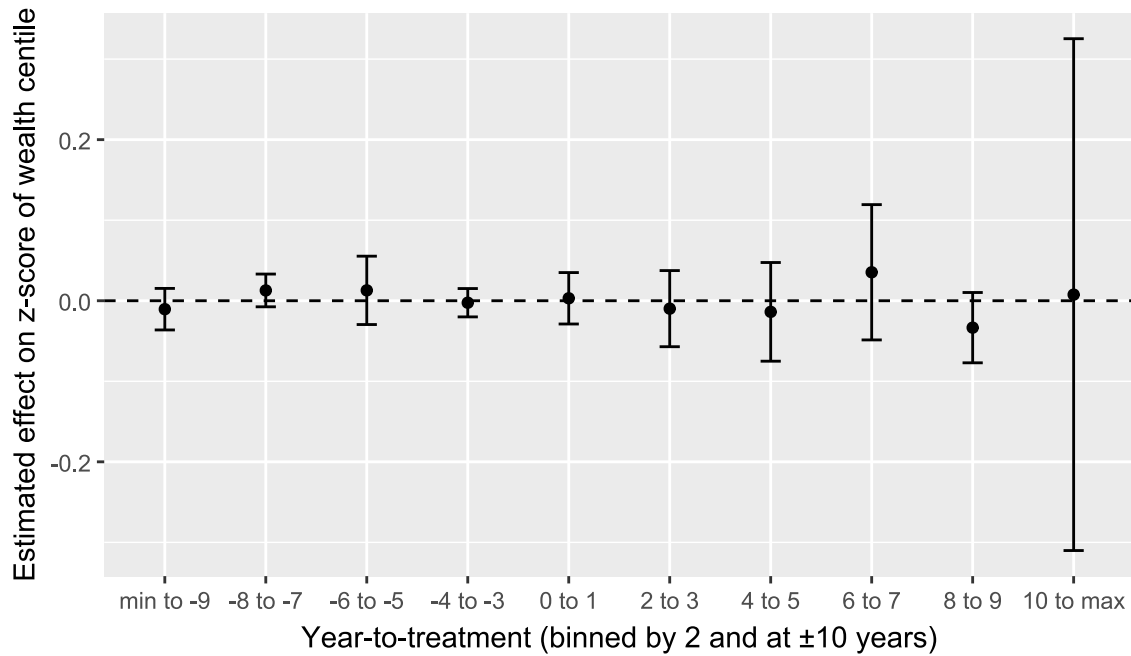
Table 5: Static treatment effect estimates on inequalities

	did2s (static)
treat_on = 1	-0.007 (0.019)
Num.Obs.	4246
R2	-0.000
R2 Adj.	-0.000
AIC	7251.5
BIC	7257.9
RMSE	0.57
Std.Errors	Corrected Clustered (cluster_uid)

- Staggered DiD: dynamic event study

Figure 9 presents the dynamic event study estimates for inequality outcomes using two year bins. Pre treatment coefficients are close to zero and statistically insignificant, supporting the plausibility of the parallel trends assumption for inequality measures. Following PA creation, estimated effects remain small and statistically insignificant across all post treatment horizons.

Figure 9: Impact on inequalities: Event-study with 2-year bins

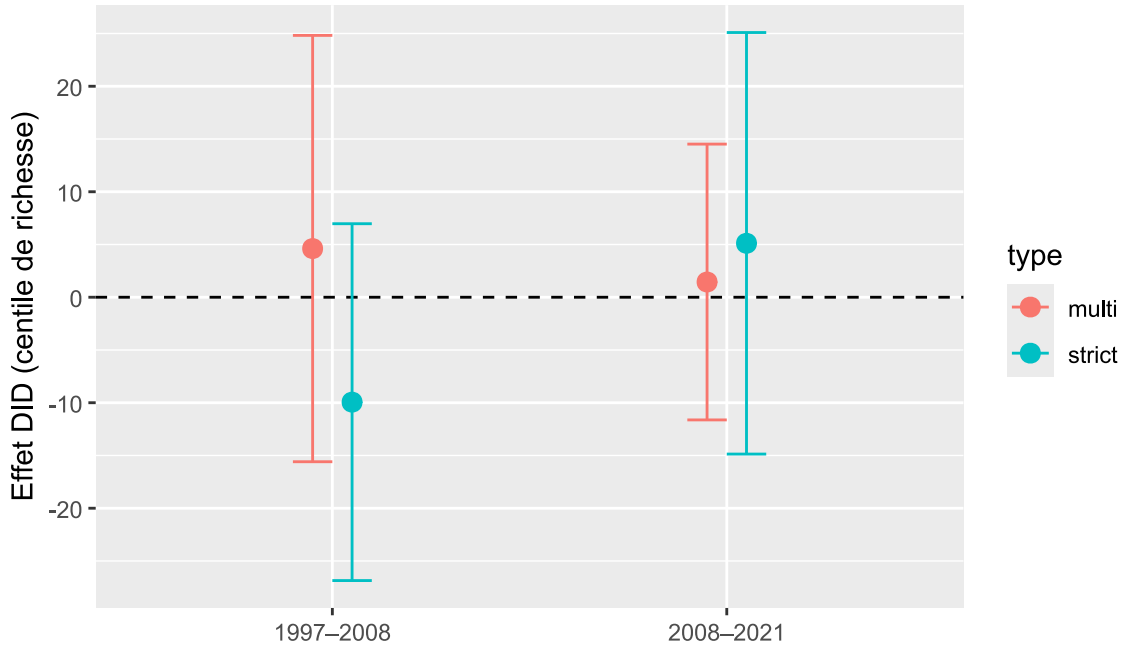


6.4 Heterogeneity

6.4.a Estimation DID 2X2

We assess whether the average effect of PA creation on household living standards differs by governance regime, distinguishing between strict PAs and multi use PAs. The 2×2 DID estimates indicate no statistically significant effect for either governance type, both in the placebo period (1997–2008) and in the treatment period (2008–2021). Point estimates differ in sign across regimes, but confidence intervals are wide and overlap zero, providing no evidence of systematic heterogeneity in average impacts by PA governance

Figure 10: Heterogeneity of impact on well-being



Notes: The graph highlights the change in household living standards before and after the PA implementation, by governance type of PA. Point estimates differ slightly between multi use and strict PAs in both periods, but confidence intervals overlap zero throughout, indicating no statistically significant differences.

staggered DID and event study specifications that allow for dynamic and cohort specific effects lead to consistent conclusions. While some coefficients are statistically significant at specific horizons, they are not stable across specifications or governance types, and do not alter the overall finding of no detectable average differential effect by PA governance regime.

6.5 Robustness

In this section, we discuss the sensitivity analyses and robustness checks that support our main results. First, to verify the robustness of our conclusions, we performed additional analyses using distances of 5 km and 15 km from the AP to define the treated group. We find that the creation of PA did not lead to significant changes in household wealth percentiles for either distance. In neither case is the interaction coefficient significant. Conversely, the temporal effect is consistently positive and significant, reflecting a general increase in wealth after 2008, regardless of the distance considered. Climatic variables are also statistically significant, and socio-demographic indicators consistently show that female-headed households have lower wealth and that wealth levels are positively correlated with the age of the household head. The stability of these results across the three distance thresholds suggests that climatic and socio-demographic dynamics largely

dominate the effect of proximity to PA. These tests showed that our conclusions are robust to the definition of treated households.

Benjamini's (Benjamini and Hochberg 1995) False Discovery rate (FDR) method showed that, prior to multiple testing, the coefficients did not show a stable trend of improvement or deterioration in the wealth index. After controlling for the risk of false positives, only a marked negative effect on the wealth index remains statistically significant. This suggests a transient effect of impoverishment.

The pseudo panel approach shows that the average wealth index of household cohorts is influenced by environmental factors. The results indicate that households located at high altitudes and in densely populated areas have a higher wealth index. Forest cover has a positive and significant effect, while slope and accessibility have a negative effect on the wealth index. In contrast, socio-demographic variables have a weakly significant effect, suggesting that spatial and environmental conditions play a more decisive role than individual household characteristics.

Rosenbaum's sensitivity test (Rosenbaum 1983) shows that the results are not robust in the face of unobserved assignment biases. In 1997, the absence of any systematic difference between households close to and far from PA before the introduction of PA confirms the hypothesis of parallel trends. In 2008, the test suggests a significant effect when assuming no bias ($\Gamma = 1$), but this quickly disappears as soon as a small unmeasured imbalance is introduced ($\Gamma > 1$). The Hodges-Lehmann intervals include zero and the p-value bounds exceed the 0.05 threshold. In 2021, the estimated effect is small and marginally significant, but it is not robust to hidden bias assumptions. Limiting ourselves to these three years, the observed wealth inequalities cannot be causally and robustly attributed to proximity to PA.

The DID estimates revealed that, on average, female-headed households have a lower wealth percentile (-2.307) than male-headed households (+0.7440). However, the interaction between the treatment and the gender of the household head does not reveal a significant difference between men and women. In other words, living near PA has no particular effect on gender inequality among household heads. Similarly, no interaction between the treatment and the age of the household head was evident, even though households headed by people aged 45-59 appeared to be significantly wealthier (2.920) than others.

7 Discussion and conclusion

This study assesses the potentially contradictory effects of PAs on the household living standards and socioeconomic inequalities among rural households in Madagascar. By relying on a pre-analysis plan and combining matching with difference in difference approaches.

Hypothesis 1 posits that, on average, PAs reduce the living standards of rural households. Our results do not support this hypothesis. Across multiple specifications, we do not detect a statistically significant average effect of PA creation on the household wealth index. Point estimates are generally negative, but confidence intervals are wide and include zero, implying that any true effect is either close to zero or below our detection capacity. A plausible explanation is that losses associated with restrictions on access to natural resources are offset, at least partially, by com-

pensatory mechanisms, ecosystem services, or adaptation strategies, even if these mechanisms remain limited or uneven. From a policy perspective, these findings underline the importance of strengthening inclusive measures, such as transparent revenue sharing from tourism or targeted transfers, to reduce the risk of vulnerability among households located near PAs

Hypothesis 2 concerns the effect of PA creation on within cluster wealth inequalities. The results show no statistically significant average impact of PAs on inequality, as measured by the Z score of the wealth index. Instead, observed changes in inequality are primarily associated with broader socio demographic and climatic dynamics. In particular, climatic shocks, as captured by drought indicators, are associated with increased inequality, suggesting that poorer households are less able to buffer adverse shocks. Household characteristics also matter: households headed by women tend to occupy lower relative positions in the wealth distribution, while those headed by older individuals tend to be better off, likely reflecting accumulated assets and social capital. These patterns point to structural drivers of inequality that operate largely independently of PA creation. In the absence of inclusive policies, conservation interventions may nonetheless interact with these dynamics by reinforcing existing asymmetries in access to opportunities, which calls for governance arrangements that explicitly address equity concerns.

Hypothesis 3 examines whether the effects of PA creation differ by governance regime. The 2x2 DID estimates do not provide evidence of statistically significant heterogeneity between strict and multi use PAs. While point estimates differ in sign across governance types, confidence intervals are wide and overlap zero, preventing any firm conclusion regarding differential average impacts. staggered DID and event study analyses lead to consistent conclusions: although some coefficients are significant at specific horizons, they are not stable across specifications and do not alter the overall finding of no detectable systematic heterogeneity by governance type. These results suggest that, in the Malagasy context, formal governance categories alone are insufficient to explain variation in socioeconomic outcomes, and that implementation quality, timing, and local context likely play a more important role

Overall, our results do not allow us to rule out that protected areas have socioeconomic impacts on rural households. However, they suggest that the widespread concern, notably in Madagascar, that PA creation substantially undermines rural living conditions is not supported at the temporal and spatial scale of our analysis. At the same time, these findings do not preclude the existence of longer term effects, nor of important sources of heterogeneity that are not captured by the dichotomous classifications used in this study. In particular, the negative effects observed at long horizons in the dynamic analysis may point to delayed or cumulative mechanisms, or to differentiated impacts across specific subsets of PAs. This calls for further work to better document long term trajectories and to move beyond coarse governance typologies in order to understand how conservation policies interact with local socioeconomic dynamics over time.

8 CRediT authorship contribution statement

Iriana Razafimahenina: conceptualization - formal analysis - Methodology - writing - original draft

Florent Bédécarrats: conceptualization - Data curation - formal analysis - Funding acquisition - investigation - Methodology - Project administration - Resources - Software - Supervision - Validation - writing - original draft -

Ingrid Dallmann: conceptualization - Methodology - Supervision - Validation - Writing - review & editing

Holimalala Randriamanampisoa: conceptualization - formal analysis - Project administration - supervision - Validation - Writing - review & editing

9 Financement

The study is performed in the framework of the BETSAKA project. The BETSAKA project is cofunded by the Development Impact Lab of the German KfW Development Bank; the Agence Française de Développement (AFD), through the PAIRES program, the French National Research Agency (ANR), and the French Research Institute for Sustainable Development (IRD).

10 Declaration of interest

One of the authors is an evaluation officer at AFD, and the BETSAKA project is funded by the Evaluation department of both AFD and KfW. While the operational departments of AFD and KfW also fund conservation projects in Madagascar and other countries, the Evaluation departments operate independently. They are committed to rigorous, unbiased studies and are supervised by independent entities within both institutions.

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